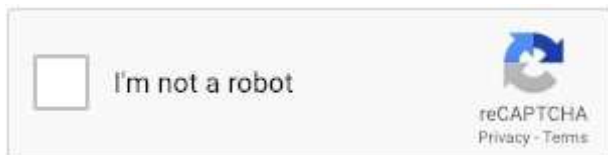


Please enter your Prolific ID below

Captcha

Please verify that you are a human by completing the CAPTCHA below.



Basic Info Block

Leave this blank



The following link will provide you with information about participating in this study: [Consent Form](#)

Please click the link to open the document (will open in a separate window) and read it carefully before agreeing to participate.

I have reviewed this form, the research staff has addressed all of my inquiries, and I agree to participate in this study.

- ☐ Agree
- ☐ Not Agree

Terminology

Vulnerability Detection – Identifying weaknesses or flaws in code, configurations, or dependencies.

Vulnerability Explanation – Describing the root cause, security impact, and potential exploitability of a vulnerability.

Vulnerability Repair – Modifying code, configurations, or workflows to remediate or prevent security issues.

Security Tasks – Activities related to improving software or system security, including detection, explanation, repair, secure design, and verification.

General-Purpose Large Language Models (LLMs) – Chat-based AI systems that generate or analyze text or code using natural-language prompts.

Examples: ChatGPT, Claude, Gemini, Llama 3/3.1 chat, Mistral Large, Mixtral chat, Cohere Command R / R+, Perplexity code chat, Replit Chat.

AI-Based Coding Tools – Machine-learning-based tools integrated into IDEs or development workflows that support coding, code review, vulnerability detection, or repair.

Examples: GitHub Copilot, Amazon CodeWhisperer, JetBrains AI Assistant, Microsoft IntelliCode AI, Tabnine, Codeium, CodiumAI, Snyk Code (DeepCode AI), Checkmarx AI, SonarQube/SonarLint AI, GitGuardian AI, Veracode Fix, Contrast Security AI.

AI Tools – In this survey, “AI tools” is an umbrella term that includes both **general-purpose Large Language Models (LLMs)** and **AI-based coding tools**. The term refers to any artificial intelligence system used to assist with security-related tasks such as vulnerability detection, explanation, or repair.

Traditional Security Tools – Established security analysis tools such as static analyzers, dynamic analyzers,

dependency scanners, CI/CD security scanners, and infrastructure security tools.

Examples: SonarQube, Semgrep, Snyk, Dependabot, AFL, Burp Suite, Trivy.

What is your primary role?

- ☐ Developer / Software Engineer
- ☐ Developer (Security Champion or Security-Heavy Responsibilities)
- ☐ Application Security Engineer / Security Analyst
- ☐ Site Reliability Engineer (SRE) / DevOps (Security-Focused)
- ☐ Security Researcher / Academic
- ☐ Student / Learner
- ☐ Other (please specify)

How often do you personally perform security-related tasks in your work?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Very often

How many years of professional coding experience do you have?

- ☐ Less than one
- ☐ One to two
- ☐ Three to five
- ☐ Six to ten
- ☐ Eleven or more

Which programming languages do you use most often?
(select all that apply)

- ☐ Python
- ☐ JavaScript or TypeScript
- ☐ Java
- ☐ C or C++
- ☐ Go
- ☐ C#
- ☐ Ruby
- ☐ Other

Which kinds of traditional security tools do you currently use to support vulnerability detection, explanation, or repair?
(Select all that apply)

- ☐ Static analysis tools (e.g., SonarQube, Fortify SCA, ESLint, FindBugs, Semgrep)
- ☐ Dependency/Library vulnerability scanning tools (e.g., Snyk, Dependabot, OWASP Dependency Check, White Source)
- ☐ CI/CD security scanning tools (e.g., GitLab SAST/DAST, GitHub Advanced Security, Jenkins security plugins, Azure DevOps security scans)
- ☐ Dynamic/runtime analysis tools (e.g., OWASP Zap, BurbSuite, Netsparker, Veracode)
- ☐ Container/infrastructure security tools (e.g., Trivy, Anchore, Clair, Aqua Security)
- ☐ Other
- ☐ None of the above

Have you ever used a general-purpose LLM (e.g., ChatGPT, Claude, Gemini) for vulnerability detection, explanation, or repair?

- ☐ Yes
- ☐ No

Which general-purpose LLMs have you used for vulnerability detection, explanation, or repair? (Select all that apply)

- ☐ ChatGPT (GPT-4, GPT-4o, GPT-4.1, GPT-3.5, etc.)
- ☐ Claude (Opus, Sonnet, Haiku)
- ☐ Gemini (Ultra, 1.5 Pro, 1.5 Flash)
- ☐ Llama (Llama 3, Llama 3.1 chat)
- ☐ Mistral / Mixtral (chat models)
- ☐ Cohere Command R / R+
- ☐ Perplexity (coding or chat models)
- ☐ Replit Chat / Ghostwriter
- ☐ I'm not sure which model I used
- ☐ Other (please specify)

Have you ever used any AI-based coding tool (e.g., Copilot, CodeWhisperer, CodeQL, or similar)?

- ☐ Yes
- ☐ No

Which AI-based coding tools have you used for vulnerability detection, explanation, or repair?

(Select all that apply)

- ☐ GitHub CodeQL
- ☐ DeepCode (Snyk Code)
- ☐ Tabnine Security
- ☐ Microsoft Security Copilot
- ☐ Amazon CodeGuru Reviewer
- ☐ GitHub Copilot
- ☐ AWS CodeWhisperer
- ☐ Google Security AI Workbench / Sec-PaLM
- ☐ CodiumAI
- ☐ Checkmarx AI Security
- ☐ Other
- ☐ None of the above

I have carefully read all questions so far. Please select “Strongly agree” to continue.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

General AI Usage for Security Tasks

For which of the following tasks do you use the AI tools you selected earlier? (LLMs + AI coding tools)? (Select all that apply.)

General-purpose LLMs

AI-based coding tools

Detecting vulnerabilities in code, configurations, or dependencies

☐☐

Explaining vulnerabilities or their root cause and impact

☐☐

Suggesting or generating code changes to repair vulnerabilities

☐☐

In what contexts have you used **AI tools** for security-related tasks? (Select all that apply)

- ☐ While coding in the IDE (inline support, autocomplete, inline analysis)
- ☐ During code review (pull requests, merge checks)
- ☐ As part of CI/CD pipelines (automated scans, build checks)
- ☐ During testing or fuzzing (runtime/dynamic checks)
- ☐ For security learning or documentation (understanding vulnerabilities, training)

☐Other (please specify)

☐ None of the above

When coding, how often do you use the AI tools you selected earlier for each of the following tasks?

	Never	Rarely	Sometimes	Often	Always
General-purpose LLMs for vulnerability detection	☐	☐	☐	☐	☐
General-purpose LLMs for vulnerability explanation	☐	☐	☐	☐	☐
General-purpose LLMs for vulnerability repair	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability detection	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability explanation	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability repair	☐	☐	☐	☐	☐

Overall, how accurate are the results from AI tools for each of the following?

	Not accurate at all	Slightly accurate	Moderately accurate	Very accurate	Extremely accurate
LLMs for vulnerability detection	☐	☐	☐	☐	☐
LLMs for vulnerability explanations	☐	☐	☐	☐	☐
LLMs for vulnerability repair	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability detection	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability explanations	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability repair	☐	☐	☐	☐	☐

When using AI tools for security tasks, how do you typically verify their output? (Select all that apply)

- ☐ Manual code review
- ☐ Run tests
- ☐ Use static/dynamic analysis tools
- ☐ Compare multiple AI answers

☐ Consult documentation

☐ I generally do not verify

☐Other

Overall, how effective do you find the AI tools for each of the following?

	Not effective at all	Slightly effective	Moderately effective	Very effective	Extremely effective
LLMs for vulnerability detection	☐	☐	☐	☐	☐
LLMs for vulnerability explanations	☐	☐	☐	☐	☐
LLMs for vulnerability repair	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability detection	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability explanations	☐	☐	☐	☐	☐
AI-based coding tools for vulnerability repair	☐	☐	☐	☐	☐

Severity & Outcomes

To ensure accuracy, please select 'Neutral' for this question.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

How severe are the most serious vulnerabilities that you have detected with the help of AI tools (LLMs + AI coding tools)?

- ☐ No real vulnerabilities (only cosmetic/style issues)
- ☐ Low severity (non-critical but worth addressing)
- ☐ Moderate severity (could cause functional or security issues in some cases)
- ☐ High severity (serious security risk if exploited)
- ☐ Critical severity (could directly compromise systems or sensitive data)
- ☐ Not sure

How risky do you consider it to rely on AI tools for security-related tasks?

- ☐ Not risky
- ☐ Slightly
- ☐ Moderately
- ☐ Very
- ☐ Extremely

For vulnerability repair, how often do AI-generated or AI-suggested fixes lead to the following outcomes?

	Never	Rarely	Sometimes	Often	Always
The fix compiles and passes tests	☐	☐	☐	☐	☐
The fix resolves the vulnerability	☐	☐	☐	☐	☐
The fix introduces a new bug	☐	☐	☐	☐	☐
The fix introduces a new security vulnerability	☐	☐	☐	☐	☐
The fix changes behavior in unintended ways	☐	☐	☐	☐	☐

Explanations & Understanding

When you use an LLM or AI-based coding tool for vulnerability detection, do you also receive or request an explanation of the vulnerability?

- ☐ The tool provides explanations automatically
- ☐ I explicitly ask for explanations
- ☐ Both (sometimes automatic, sometimes requested)
- ☐ I typically do not get explanations

How confident are you in AI-generated vulnerability explanations or fixes?

- ☐ Not confident
- ☐ Slightly
- ☐ Moderately
- ☐ Very
- ☐ Extremely

How clear and understandable are AI-generated explanations of vulnerabilities?

	Not clear at all	Slightly clear	Moderately clear	Very clear	Extremely clear
Explanations from LLMs	☐	☐	☐	☐	☐

Not clear at all

Slightly clear

Moderately clear

Very clear

Extremely clear

Explanations from AI-based coding tools

☐

☐

☐

☐

☐

How often are AI-generated vulnerability explanations helpful for you in practice?

Never

Rarely

Sometimes

Frequently

Always

Explanations from LLMs

☐

☐

☐

☐

☐

Explanations from AI-based coding tools

☐

☐

☐

☐

☐

What challenges have you encountered with AI-generated vulnerability explanations? (Select all that apply.)

- ☐ Explanations were too vague or generic
- ☐ Explanations were too technical or jargon-heavy
- ☐ Explanations lacked sufficient context about the codebase
- ☐ Explanations overstated or understated the severity
- ☐ Explanations were incorrect or misleading
- ☐ Explanations recommended insecure patterns
- ☐Other (please specify)

Perception Comparison: LLMs vs AI-Based Coding Tools

How likely are you to continue or expand your use of **AI tools** in the future for

	Extremely unlikely	Somewhat unlikely	Neither likely nor unlikely	Somewhat likely	Extremely likely
detecting vulnerabilites	☐	☐	☐	☐	☐
explaining vulnerabilities	☐	☐	☐	☐	☐
repairing vulnerabilities	☐	☐	☐	☐	☐

For each of the following security tasks, which type of AI tool do you generally find performs better?

	General-purpose LLMs	AI-based coding tools	About the same	Depends on the situation	Not sure
Vulnerability detection (more accurate results)	☐	☐	☐	☐	☐

	General- purpose LLMs	AI-based coding tools	About the same	Depends on the situation	Not sure
Vulnerability repair (more trustworthy fixes)	☐	☐	☐	☐	☐
Vulnerability explanation (clearer reasoning)	☐	☐	☐	☐	☐

Can you describe a situation where an AI tool gave a misleading or incorrect security recommendation?

Traditional Tools vs AI Tools

Select Strongly Disagree from the options below.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

How often do you use traditional (non-AI) security tools in addition to AI tools when working on vulnerabilities?

	Never	Rarely	Sometimes	Often	Always
For vulnerability detection	☐	☐	☐	☐	☐
For vulnerability explanation	☐	☐	☐	☐	☐
For vulnerability repair	☐	☐	☐	☐	☐

When working on vulnerabilities, how often do you prefer traditional tools over AI tools?

	Never	Rarely	Sometimes	About half the time	Most of the time	Always
For vulnerability detection	☐	☐	☐	☐	☐	☐
For vulnerability explanation	☐	☐	☐	☐	☐	☐
For vulnerability repair	☐	☐	☐	☐	☐	☐

Which traditional tools do you most often combine with AI tools for each of the following?

	Traditional Static Analyzers	Dependency/Supply Chain Tools	Dynamic Analysis & Fuzzing Tools	Container & Infrastructure Security Tools
Detection	☐	☐	☐	☐
Explanation	☐	☐	☐	☐
Repair	☐	☐	☐	☐

Follow-up

Is there anything else you'd like to share about using AI tools for security tasks?

Thank you for completing this survey! We appreciate your time and thoughtful responses. We occasionally conduct follow-up or related studies on this topic. If you would like to be contacted about future research opportunities or to be entered into the raffle, please indicate below. (If you select “Yes,” please provide your email address. Your contact

information will be stored separately from your survey responses to maintain confidentiality.)

- ☐ Yes, I am interested in being contacted for future surveys and entering into the raffle.

- ☐ No future contact — I am only providing my email to enter the raffle.

- ☐ No, I do not wish to be contacted or entered into the raffle.

If you know someone who might also be eligible and interested in participating, please feel free to share the survey link or flyer with them. Do not send us anyone's personal information, simply share the public link so they can review the consent form and decide whether to participate. Thank you again for your contribution; your responses are valuable and greatly appreciated!

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